Prajwal Singh

Backend Engineer | LLM Applications & AI Systems

Bengaluru, India | +91 7398314111 | prajwalsingh0605@gmail.com
linkedin.com/in/iprajwalsingh7 | github.com/iprajwalsingh7

Summary

Backend engineer building production trading systems and LLM-powered workflow automation using Java, Spring Boot, Redis and Lua. Delivered natural-language bot creation across 24 trading strategies, trailing stops and tiered exits. Enthusiastic about applied AI and actively developing Python and machine learning skills to build reliable AI applications.

Experience

Option Circle | Junior Software Engineer (Backend) Jun 2026 – Present
Bengaluru, India

- Shipped tiered profit-taking across **4 brokers plus paper trading**: TradeStation, TastyTrade, Tradier and Webull, enabling configurable partial exits instead of all-or-nothing position closure.
- Implemented tier sizing from original holdings, preserving multi-leg spread ratios and accounting for working orders to prevent overselling; closed all remaining contracts at the final tier.
- Coordinated broker fills, partial fills, cancellations and retries with Redis order-to-tier state, duplicate-message guards and atomic fill claims to prevent duplicate sell records.
- Built a **Redis Lua trailing-stop engine** supporting profit and loss trailing, with atomic per-tick peak tracking and exit evaluation; routed close requests through Kafka.
- Added shared close locks, broker-stage lock extension and trigger-time checks to suppress duplicate exit requests while allowing failed closes to retry.

Option Circle | Backend Engineering Intern Jan 2026 – May 2026
Bengaluru, India

- Built a production **LLM parser and conversion layer** using Gemini, Java, Spring Boot and embedded LuaJ; converted natural-language instructions into validated workflow nodes and edges across **24 strategies**, then persisted and scheduled bots through Bot-Engine.
- Added server-side checks for required fields and allowed values, plus build-time checks against UI and engine definitions to detect configuration drift before deployment.
- Integrated Gemini through non-blocking **Spring WebFlux** clients; generated the structured specification and explanation in **one model request**.

Projects

Cooked | LLM-Powered Profile Aggregation GitHub

- Built a Java/Spring Boot service combining GitHub, Reddit and Spotify data with Gemini-generated responses; implemented Spotify OAuth 2.0 token exchange, API error handling and JSON parsing.

TickPulse | Real-Time Market Alerts GitHub

- Built Redis Streams ingestion and Lua rule evaluation with Server-Sent Events, PostgreSQL alert history and Docker; added Testcontainers integration tests for Redis state transitions.

Technical Skills

Languages: Java, Python, SQL, Lua, Go

AI & Backend: LLM API integration, structured outputs, Gemini API, Spring Boot, Spring WebFlux, Spring Data JPA, REST APIs, OAuth 2.0

Data & Messaging: Redis, Redis Streams, Apache Kafka, PostgreSQL, MySQL, MongoDB, Elasticsearch

Tools & Engineering: Docker, Git, Maven, Linux, Postman, Testcontainers, concurrency, idempotency, event-driven architecture, distributed systems

Education

MVJ College of Engineering, Bengaluru 2022 – 2026
B.E., Information Science and Engineering | CGPA: 8.29/10